

Lab 06: Departmental Subnetting

Subnetting for IT · HR · Finance Departments

Lab #6

Topic: Subnetting &
VLANs

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Divide a 192.168.100.0/24 network into three departmental subnets.
- Configure each subnet on a separate router interface.
- Assign hosts in each department to the correct subnet.
- Verify inter-department routing through the central router.
- Relate to Network+ objective: 1.4 – Subnetting and addressing.

2. Subnetting Design

The address block 192.168.100.0/24 is divided into three /26 subnets providing 62 usable hosts per department:

Department	Subnet	Mask	Gateway	Host Range	Hosts
IT	192.168.100.0/26	255.255.255.192	192.168.100.1	.2 – .62	62
HR	192.168.100.64/26	255.255.255.192	192.168.100.65	.66 – .126	62
Finance	192.168.100.128/26	255.255.255.192	192.168.100.129	.130 – .190	62

3. Network Topology

One central router (R1) has three FastEthernet interfaces — one per department. Each department has a switch and 2–3 PCs. The router performs inter-department routing.

Device	Department	Interface	IP Address	Mask
R1	IT	Fa0/0	192.168.100.1	255.255.255.192
R1	HR	Fa0/1	192.168.100.65	255.255.255.192
R1	Finance	Fa1/0	192.168.100.129	255.255.255.192
IT-PC0	IT	—	192.168.100.10	255.255.255.192
HR-PC0	HR	—	192.168.100.70	255.255.255.192

FIN-PC0	Finance	—	192.168.100.140	255.255.255.192
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4. Step-by-Step Configuration

4.1 Router R1 – Three Interfaces

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R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 192.168.100.1 255.255.255.192
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# interface FastEthernet0/1
R1(config-if)# ip address 192.168.100.65 255.255.255.192
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# interface FastEthernet1/0
R1(config-if)# ip address 192.168.100.129 255.255.255.192
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# exit
R1# write memory

```

5. Verification

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IT-PC0> ping 192.168.100.70      ! IT → HR
IT-PC0> ping 192.168.100.140    ! IT → Finance
HR-PC0> ping 192.168.100.140    ! HR → Finance
R1# show ip route                ! Verify all three connected routes
R1# show ip interface brief      ! All interfaces should be up/up

```

6. Key Takeaways

- A /26 subnet provides 64 total addresses, 62 usable — ideal for small departments.
- Subnetting allows traffic isolation; combined with ACLs, departments can be kept separate.
- A router-on-a-stick or multi-interface router can connect multiple subnets without needing separate physical routers.
- Giving each department its own subnet makes IP management, security policy, and troubleshooting much easier.
- Always document your subnet plan before configuring — mistakes in subnet math affect all devices.